
Invocation-Level Reliability of Tool-Using Agents

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Abstract

1 An early tool-call error can corrupt every downstream input even when an agent
2 subsequently applies the right local rule. We isolate this effect with matched
3 evaluations under a correct baseline history and the model’s own free-running
4 history. For tasks of depth d , their pooled invocation rates $\bar{p}_d$ and $\bar{g}_d$ define a fit-
5 free relative propagation loss $L_d = 1 - \bar{g}_d/\bar{p}_d$. Across five open-weight models on
6 controlled routing tasks, the two 7–8B models lose about 68% of baseline capability
7 by depth 6; stronger models mostly remain at ceiling. Our main result is a construct-
8 validity condition for trajectory scoring. Let Y_k^* be the canonical next target after a
9 divergence and H_k the agent-observable history. If Y_k^* has conditional guessing
10 probability at most ϵ_k given H_k , then any fixed-gold scorer credits the agent with
11 probability at most ϵ_k , regardless of its local competence. Gold-agreement severity
12 is therefore driven to a scorer-imposed boundary, and observed reconvergence
13 cannot measure behavioral recovery. **Fixed reference alone is not sufficient:** the
14 result applies when the canonical target becomes unreachable or unpredictable,
15 and excludes reconstructible targets and evaluators that accept alternative states
16 or milestone paths. Our hidden-constant tasks instantiate the condition: 0 of
17 869 corrupted-context calls match gold, and 0 of 580 opportunities reconverge.
18 Conditional-on-state replay instead asks whether a call correctly continues from
19 the state the agent actually holds; it yields interior severity estimates of 0.149
20 and 0.316 and requires no new model queries. Alternative absolute-gap and error-
21 amplification summaries give the same qualitative depth trend.

1 Introduction

23 Language model agents interact with external systems by emitting structured *tool calls*: each call
24 names a function (e.g. `get_weather`, `query_database`) and fills its required arguments. A call
25 can fail by selecting the wrong function (*selection error*) or by providing incorrect argument values
26 (*argument error*). In a multi-step task where each call’s output feeds into the next, a single wrong call
27 early in the chain corrupts the inputs to every downstream call, even if the model reasons correctly
28 from that point on.

29 Many benchmarks report task success or agreement with a canonical call, without separately mea-
30 suring selection, arguments, and the downstream effect of an earlier error. State-based benchmarks
31 allow alternative solutions, and process benchmarks annotate individual steps, but neither directly
32 supplies the matched baseline/free-running contrast studied here. These distinctions matter because
33 local incompetence, state corruption, and ordinary later-step difficulty require different interventions.

34 We introduce a protocol that makes this distinction. A call is *correct* when it selects the appropriate
35 tool and supplies the arguments required by the scoring regime. We measure it under two matched
36 conditions: a *baseline* condition with a guaranteed-correct history and a *free-running* condition
37 with the model’s own prior outputs. Their gap measures the loss associated with accumulated state
38 corruption while controlling for the prompts and later-step difficulty present in the baseline arm.

The study began as an empirical comparison and exposed a construct-validity problem. When divergence makes the canonical next target unpredictable from everything the agent observes, gold agreement measures target reachability rather than the competence of the next call. This is not a property of every fixed-reference benchmark: it depends on an explicit unreachable-target condition. We state that condition, give a counterexample, and separate the resulting scorer-imposed boundary from the model’s behavior.

Contributions.

1. A disaggregated measurement protocol that separately scores tool selection and argument correctness under matched baseline and free-running conditions, with a routing-task design where selection errors, not just argument errors, propagate.
2. A formal unreachable-target condition under which canonical gold agreement is bounded by the target’s conditional guessing probability, making gold-based severity and reconvergence scorer-dominated rather than behavioral.
3. A remedy, conditional-on-state scoring, where each call is judged against what would be correct given the value the model actually holds: implemented and applied retrospectively to cached completions at zero marginal cost.
4. Robustness summaries beyond L_d (absolute excess error and error amplification), plus per-model diagnostics for rule discrimination and error position.

2 Related Work

Toolformer [1], ReAct [2], and Gorilla [3] established tool-augmented agents. ToolLLM [4], Stable-ToolBench [5], API-Bank [6], and BFCL [7] emphasize task success or agreement with reference calls. RoTBench [9] and MTU-Bench [10] separate selection from argument accuracy; FuncBenchGen [11] generates fresh dependency-graph tasks and reports stale-value failures. Our distinct estimand is the matched change in invocation correctness when gold history is replaced by the same model’s history.

Not all agent evaluation uses one canonical trajectory. τ -bench [8] grades final environment state; AppWorld [16] uses state-based unit tests that permit multiple solutions; ToolSandbox [17] evaluates intermediate and final milestone DAGs over arbitrary trajectories. These methods avoid the unreachable-canonical-target failure but primarily measure outcome or milestone completion, not local post-divergence competence. Section 5.3 states the narrower condition under which canonical-call scoring becomes scorer-dominated.

Process-oriented work is complementary. AgentProcessBench [18] provides human step labels and an explicit propagation rule, while FALAT [19] traces dependencies to separate decisive errors from inherited downstream failures. Relign [12], Hammer [13], and representation-level detection [14] instead reduce or detect the production of an incorrect call. We intervene on history to quantify propagation and compare gold-agreement with conditional-on-state scoring.

3 Method

3.1 Definitions

A task of dependency depth d contains calls indexed by position $k \in \{1, \dots, d\}$. For task i , let $B_{i,k}$ and $F_{i,k}$ indicate invocation correctness in the baseline and free-running arms. With N_d tasks at depth d , the reported depth-level estimands are

$$\bar{p}_d = \frac{1}{N_d d} \sum_{i=1}^{N_d} \sum_{k=1}^d B_{i,k}, \quad \bar{g}_d = \frac{1}{N_d d} \sum_{i=1}^{N_d} \sum_{k=1}^d F_{i,k}. \quad (1)$$

- **Pooled baseline correctness** $\bar{p}_d$: invocation accuracy with a guaranteed-correct prior history.
- **Pooled free-running correctness** $\bar{g}_d$: invocation accuracy with the model’s own prior outputs.
- **Relative propagation loss** $L_d := 1 - \bar{g}_d/\bar{p}_d$. We also report the absolute gap $A_d = \bar{p}_d - \bar{g}_d$ and error amplification $E_d = (1 - \bar{g}_d)/(1 - \bar{p}_d)$ (Appendix A).

L_d is descriptive and invocation-pooled; it is not the stepwise quantity in the Markov recurrence below. Calls from one task are dependent, so inferential intervals must resample matched tasks jointly rather

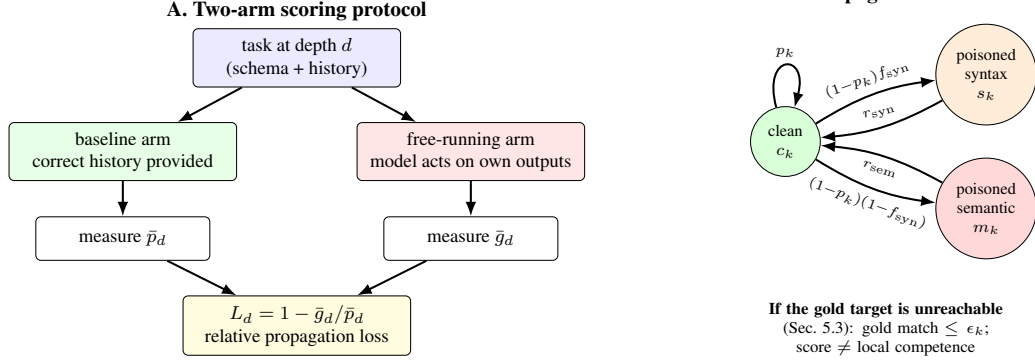


Figure 1: (A) Matched evaluation at task depth d : pooled baseline and free-running invocation rates define L_d . (B) A separate step-indexed propagation model. Under the unreachable-target condition of Section 5.3, canonical gold agreement is bounded by target predictability rather than local competence.

85 than treat $N_d d$ calls as independent. The conclusions emphasized here are the paired point-estimate
 86 pattern, the directly observed corrupted-context counts, and the scorer theorem; Appendix A exposes
 87 the aggregate counts recoverable from the paper.

88 3.2 A propagation model with recovery

89 For interpretation only, we model step-indexed context as clean, poisoned-by-syntax, or poisoned-
 90 by-semantics (Figure 1B). Let p_k be correctness at position k under clean history and $f_{syn}(k)$ the
 91 syntactic share of new errors. With origin-specific recovery rates r_{syn}, r_{sem} and poisoned-state
 92 correctness $\rho_k = (1 - \pi)p_k$, occupancies evolve as

$$c_{k+1} = c_k p_k + s_k r_{syn} + m_k r_{sem}, \quad s_{k+1} = c_k (1 - p_k) f_{syn}(k) + s_k (1 - r_{syn}), \quad (2)$$

$$m_{k+1} = c_k (1 - p_k) (1 - f_{syn}(k)) + m_k (1 - r_{sem}). \quad (3)$$

93 The model-implied step accuracy is $g_k^{\text{model}} = c_k p_k + (s_k + m_k) \rho_k = p_k (1 - \pi x_k)$ for poisoned
 94 mass $x_k = s_k + m_k$. We fit π, r_{syn}, r_{sem} hierarchically (PyMC, NUTS) as a cautionary diagnostic,
 95 not as the estimator of L_d . Section 5.3 explains why gold-agreement data do not identify the intended
 96 behavioral interpretation of these parameters under an unreachable target.

97 3.3 Task design

98 Each task is a hidden dependency graph over deterministic executable functions, giving exact
 99 ground truth per call and direct control of depth ($\{1, 2, 4, 6, 8\}$). Tool outputs lie in a size- M space
 100 ($M = 100,000$) and take the form $(a \cdot \arg + b) \bmod M$, with per-tool constants omitted from the
 101 schema. Section 5.3 does not infer unpredictability from secrecy alone; it states the conditional
 102 target-predictability assumption required for the bound.

103 **Routing tasks (primary)**: the correct next tool is a parity rule applied to the incoming value, not a
 104 pre-announced order, so a wrong choice changes what is correct downstream. Selection is scored
 105 *conditionally*, against the tool correct given the value the model actually holds, with divergence
 106 from gold recorded separately, so applying the rule correctly to a corrupted input is not misclassified
 107 as selection failure. **Linear tasks (control)**: order announced, argument copied verbatim. Pilot
 108 runs were degenerate ($\bar{p}_d = \bar{g}_d = 1.000$ on two models), and the full null arm was not executed
 109 (Section 7).

110 3.4 Protocol and suite

111 We score selection against the tool appropriate for the value in the model’s actual state and, for the
 112 primary gold-agreement analysis, score arguments against the canonical trajectory. Digit-only JSON
 113 strings are coerced to integers, so formatting is not a value error. Errors are syntactic (parse/schema

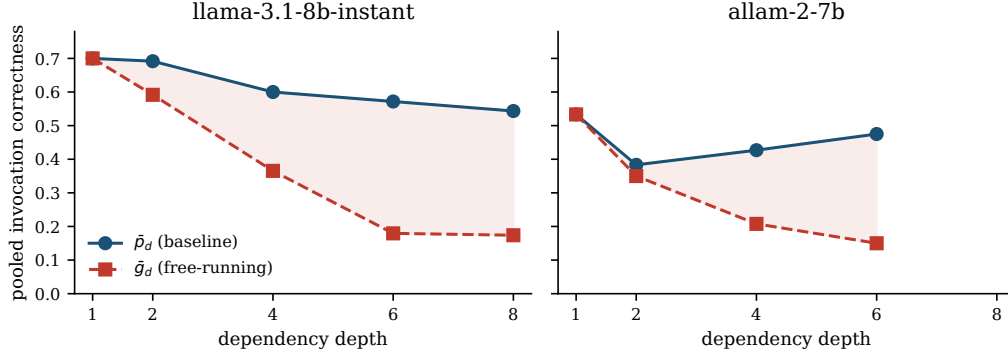


Figure 2: Pooled baseline correctness $\bar{p}_d$ versus pooled free-running correctness $\bar{g}_d$. The absolute gap widens through depth 6 and then plateaus for llama-3.1-8b; depth 1 is identical by construction.

failure) or semantic (executes but is wrong). Both arms use the same within-step retry budget. Conditional-on-state replay replaces the argument target with the argument implied by the model’s actual state, making the selection and argument criteria symmetric.

We evaluated five open-weight models available through one hosted API: llama-3.1-8b-instant, allam-2-7b, qwen3.6-27b, gpt-oss-120b, and llama-3.3-70b-versatile. Greedy decoding, fixed seeds, and caching by (model, calling mode, exact prompt) make each benchmark prompt deterministic. Provider constraints produced unequal nested sample sizes and prevented balanced family, proprietary-model, and calling-mode comparisons; we draw no claims along those axes.

Models run **nested prefixes** of one task suite, sized per model to its own measured token allowance. Task identity depends only on (depth, k , seed), never on how many tasks were requested, so cross-model contrasts on the shared prefix remain exactly paired while each model’s own estimate uses its full n .

4 Data Collected

All figures come from a dataset frozen at 14:09 on 2026-08-17; no missing model–depth cell is extrapolated.

Table 1: Per-model sample sizes at the freeze.

Model	tasks	d_1	d_2	d_4	d_6	d_8	status
llama-3.1-8b-instant	273	60	60	65	65	23	only model at depth 8
allam-2-7b	225	60	60	65	40	–	usable
gpt-oss-120b	108	26	26	28	28	–	usable
qwen3.6-27b	98	26	26	28	18	–	usable
llama-3.3-70b-versatile	49	12	12	13	12	–	usable, thin
gpt-oss-20b	0	–	–	–	–	–	excluded

gpt-oss-20b completed zero tasks after pilot calls consumed its daily allowance and is excluded from every empirical claim. Provider constraints explain the unequal sample sizes but are not treated as scientific findings.

5 Results

5.1 Propagation grows through depth 6 and exceeds baseline decay

Figure 2 and Table 2 show the core measurement. On llama-3.1-8b, $\bar{p}_d$ declines from 0.700 to 0.543 (22% relative), while $\bar{g}_d$ declines to 0.174 (75%). L_d rises sharply through depth 6 and is essentially unchanged at depth 8 (0.686 vs. 0.680); we therefore claim growth followed by a plateau, not strict monotonicity or significant adjacent-depth differences.

Table 2: Invocation-pooled depth estimands for the two high- n models. Intervals are the reported 89% intervals from the frozen paired analysis.

llama-3.1-8b-instant					
depth d	$\bar{p}_d$	$\bar{g}_d$	L_d [89% interval]	calls	
1	0.700	0.700	0.000 [0.000, 0.000]	60	
2	0.692	0.592	0.145 [0.082, 0.222]	120	
4	0.600	0.365	0.391 [0.311, 0.474]	260	
6	0.572	0.179	0.686 [0.607, 0.766]	390	
8	0.543	0.174	0.680 [0.574, 0.772]	184	
allam-2-7b					
1	0.533	0.533	0.000 [0.000, 0.000]	60	
2	0.383	0.350	0.087 [0.022, 0.163]	120	
4	0.427	0.208	0.514 [0.429, 0.600]	260	
6	0.475	0.150	0.684 [0.624, 0.745]	240	

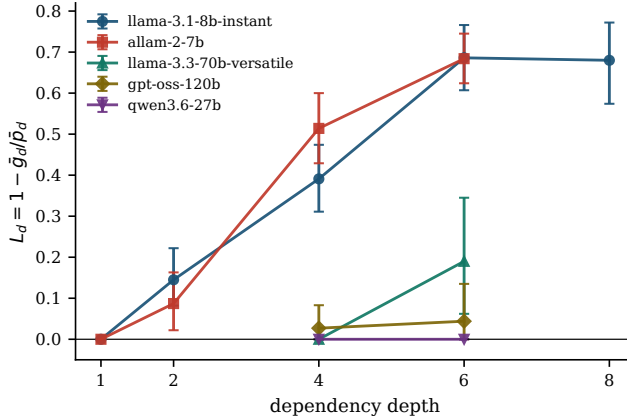


Figure 3: L_d versus depth for all usable models (89% intervals). Three models remain near ceiling; llama-3.3-70b-versatile shows a positive depth-6 estimate.

138 $L_1 = 0$ **exactly on every model**: both arms build an identical depth-1 prompt and share one
139 cached response, providing an implementation invariant. At depth 6, llama-3.1-8b has identical
140 baseline/free error counts at the first position (21/21), after which free-running errors rise while
141 baseline errors remain approximately flat. The alternative aggregate summaries in Appendix A show
142 the same through-depth-6 increase without dividing by baseline accuracy.

143 The lag association $\Delta_1 = P(k+1 \text{ correct} \mid k \text{ correct}) - P(k+1 \text{ correct} \mid k \text{ wrong})$ is positive for
144 every model with errors (+0.32 to +1.00). It is descriptive rather than causal; its second term is zero
145 under canonical gold agreement because of the unreachable-target mechanism below.

146 5.2 Ceiling models, and where in a chain a model errs

147 Three models sit at or near ceiling ($\bar{p}_d = 0.865$ –1.000; Figure 3), so $L_d \approx 0$ is primarily a statement
148 about task difficulty. On llama-3.3-70b-versatile at depth 4, 6 of 7 errors occur at a terminal
149 position, leaving no downstream opportunity. At depth 6 its estimate is $L_6 = 0.190$ [0.062, 0.345],
150 but the sample is thin. gpt-oss-120b has one fresh error, and qwen3.6-27b has zero errors across
151 298 free-running calls; neither supports a robustness claim.

152 An error at the terminal position cannot propagate, so a fixed-depth aggregate mixes error fre-
153 quency with the remaining opportunity for downstream damage; $L_1 = 0$ is the limiting case. For
154 llama-3.1-8b, terminal-error shares are 0.714 at $d = 2$ and 0.197 at $d = 6$, versus uniform-position
155 baselines of 0.500 and 0.167. For llama-3.3-70b-versatile, the 6/7 terminal share at $d = 4$
156 is suggestive but extremely uncertain. Terminal-error share should therefore accompany L_d , and
157 cross-model comparisons require comparable position distributions.

5.3 When canonical gold agreement ceases to measure behavior

Let H_k denote everything available to the agent before a post-divergence call, and let Y_k^* be the canonical gold target used by the scorer. Define its conditional guessing probability

$$\epsilon_k(H_k) = \max_y P(Y_k^* = y \mid H_k). \quad (4)$$

Proposition (unreachable-target bound). For any possibly randomized agent whose prediction $\hat{Y}_k$ depends only on H_k ,

$$P(\hat{Y}_k = Y_k^* \mid H_k) \leq \epsilon_k(H_k). \quad (5)$$

Consequently, if q_k^{gold} is correctness in corrupted context and $p_k > 0$ is clean-context correctness, gold-defined severity satisfies $\pi_k^{\text{gold}} = 1 - q_k^{\text{gold}}/p_k \geq 1 - \mathbb{E}[\epsilon_k]/p_k$. Across recovery opportunities $\mathcal{K}$, the probability of any chance reconvergence is at most $\sum_{k \in \mathcal{K}} \mathbb{E}[\epsilon_k]$.

Proof. Conditional on H_k , sum over candidate outputs: $P(\hat{Y}_k = Y_k^* \mid H_k) = \sum_y P(\hat{Y}_k = y \mid H_k) P(Y_k^* = y \mid H_k) \leq \max_y P(Y_k^* = y \mid H_k)$. The recovery statement is the union bound. $\square$

The proposition is about *construct validity*: when ϵ_k is negligible, gold agreement cannot distinguish a locally competent continuation from arbitrary behavior. It does not say the operational statistic is numerically unidentified; rather, the statistic is pinned near a boundary that no longer represents the intended behavioral severity.

Instantiation in this suite. After divergence, canonical arguments depend on the hidden-constant output associated with the gold state. Under the generator assumption that this target is conditionally uniform over $M = 100,000$ values, $\epsilon_k = 10^{-5}$. We observe 0 gold matches among 869 corrupted-context calls and 0 canonical reconvergences among 580 opportunities. Even a pure-guess process has expected reconvergences $580/M = 0.0058$, with probability of any such event at most 0.0058. Zero therefore carries essentially no information about whether the agent can apply the local routing/copy rule in its actual state.

If one nevertheless substitutes the scorer-implied boundary $\pi = 1$ and zero canonical recovery into the step model, it collapses to $g_k^{\text{model}} = c_k p_k$ and $c_k = \prod_{j < k} p_j$. A hierarchical fit to pooled aggregates returned severity posteriors of 0.92 and 0.73 despite zero observed post-divergence gold matches; $\hat{R} \leq 1.0005$, ESS $\geq 5,823$, and zero divergences only show that the misspecified posterior was explored successfully.

Counterexample and scope. Fixed reference is not sufficient for the proposition. If Y_k^* is reconstructible from the observable history, independent of the corrupted value, or reachable through reset/re-query actions, ϵ_k may be large and canonical recovery can be meaningful. Final-state unit tests and milestone-DAG evaluators can also accept alternative valid trajectories. Our claim is therefore restricted to canonical scoring under an empirically or structurally small ϵ_k ; it is not a blanket claim about execution match, AST match, or fixed references.

5.4 A remedy: conditional-on-state scoring

Credit a call when it *correctly continues from the value the model actually holds*, not when it matches the canonical state. Routing selection was already computed conditionally; replay adds the argument-side counterpart. Because every raw completion was cached, this required 881 cache reads and no new model queries.

Clean-context calls score identically under both rules (0.658 vs. 0.658), an implementation invariant. Conditional severity is 0.149 [0.021, 0.268] for llama-3.1-8b and 0.316 [0.066, 0.503] for llama-2-7b. Thus state corruption is associated with genuine loss of local rule application, but far less than the boundary value assigned by canonical gold agreement. Evidence is limited to two models and a weak copy-argument competence test.

Trade-off. Conditional scoring requires an executable transition model for the agent’s actual state and does not imply end-task success. Canonical gold, conditional-on-state, milestone, and final-state scores answer different questions; none should be presented as a universal replacement.

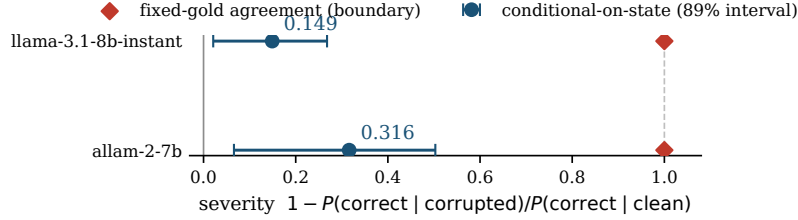


Figure 4: Gold-agreement severity is pinned at its boundary, whereas conditional-on-state replay yields interior estimates (reported 89% intervals). The contrast is a change of estimand, not an improved fit to the same target.

203 5.5 Rule-following is measurable directly, and aggregates hide it

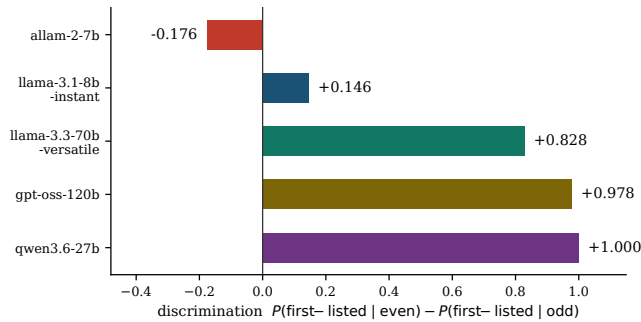


Figure 5: Discrimination, defined as $P(\text{first-listed} \mid \text{even}) - P(\text{first-listed} \mid \text{odd})$, per model. Order matches the independently established reliability order of Section 5.1.

204 Define **discrimination** as $P(\text{first-listed} \mid \text{ref even}) - P(\text{first-listed} \mid \text{ref odd})$: 0 means the rule is
 205 ignored, +1 perfect application, negative means *anti*-correlated. Overall first-listed rates (0.478–
 206 0.497, four of five models) are indistinguishable and consistent with no position bias whatever; the
 207 *same data*, conditioned on parity, resolves discrimination across the full range (Figure 5), and the
 208 order matches reliability established independently in Section 5.1. **Discrimination is a per-model**
 209 **diagnostic of whether a model performs the task at all, invisible to any metric aggregated over**
 210 **the conditioning variable.**

211 Three qualitatively different relationships, not one spectrum: qwen3.6-27b *derives* the rule (+1.000,
 212 zero errors in 298 steps, explicit reasoning trace); llama-3.1-8b *weakly follows* it (+0.146, $z=4.7$);
 213 allam-2-7b is *anti-correlated* (−0.176, $z=-5.0$): it picks the first-listed tool *more* on odd refs
 214 (0.774) than even (0.597), where its accuracy is worst (0.223 vs. 0.595). allam-2-7b’s propagation
 215 loss is therefore not “a weak model that errs more” but a model *not performing the routing task*.

216 5.6 Two negative results

217 **Untestable, not underpowered:** a hypothesised error-composition shift along the chain cannot be
 218 tested here because **389 of 389** clean-context errors are selection errors: on a copy-argument task
 219 there are only two ways to fail cleanly, mis-apply the parity rule (hard) or mis-transcribe a verbatim
 220 number (models never do). **Unresolved by design:** the suite offers two 2-point scale contrasts rather
 221 than 3–4, not comparable in kind (llama 8B→70B dense, $8.75\times$; gpt-oss 20B→120B, $6\times$ total
 222 but $\approx 1.4\times$ active params). We report both axes and draw no scale conclusion, since the larger
 223 models sit at ceiling on these tasks.

6 Pipeline Validation, and What Simulation Cannot Establish

Before spending API budget we validated the full pipeline against six simulated policies with known ground truth, eliminating seven harness artifacts (retry-budget asymmetry, selection errors mis-attributed under poisoning, a contaminated syntactic-share estimator, among others). Two further artifacts were not coding errors, which is what makes them instructive: comparing carried value against *expected argument* rather than *previous gold output* is correct on a copy-argument task and silently wrong under any transform; and the simulated policies recovered by reading ground truth directly, out of band, through a channel no real model has. Correct code, correct configuration, invalid measurement. The second artifact is why the recovery channel was never estimable from observable data in any version of this pipeline.

Three principles follow, nested by scope: **(1) simulated validation** certifies only the channels it exercises and silently passes the ones it does not (our mock backend read state from stashed context rather than the message history, so an agent loop that never threaded observations into the conversation passed the entire simulated suite); **(2) a parser or scorer can silently flip the direction of a result**, not just its precision: replaying cached completions through our pre-fix extractor, **319/394 (81.0%)** of qwen3.6-27b’s responses would have failed to parse; **(3) every constant estimated from one model or assumed for convenience should be verified per-model**: a shared output-token constant hid qwen’s true cost ($6.6\times$ higher); an assumption that parity was incidental hid allam’s anti-correlation; per-depth pooling hid where each model’s errors cluster.

This establishes that the estimation and aggregation pipeline recovers configured parameters. It does **not** establish the prompt-construction and observation-passing path, and no out-of-band simulation can.

7 Limitations and Scope

This study uses synthetic integer-argument tasks, giving exact state transitions at the cost of ecological validity. The strongest empirical findings are carried by two 7–8B models; three stronger models are largely at ceiling. The unreachable-target proposition is general only through its explicit ϵ_k condition, while the empirical claim that our generator makes ϵ_k negligible depends on its hidden-state construction. Validation on natural-language arguments and stateful environments such as ToolSandbox or AppWorld is required before claiming prevalence in deployed agents.

Four designed conditions were not executed: a linear-task null control, transformed arguments, randomized presentation order, and a calling-mode ablation. These are material threats, not cosmetic extensions. In particular, allam-2-7b’s anti-correlation may depend on the parity/order presentation, and copy arguments do not test semantic argument construction. The unequal nested sample sizes preclude scale or family comparisons. The reported intervals require the task-level artifact for reproduction; call-level binomial intervals would be invalid because steps within a task are dependent.

8 Conclusion

Matched baseline/free-running evaluation reveals large propagation on two small models: at depth 6, pooled free-running correctness is about 68% below each model’s baseline rate. Absolute excess error and error amplification show the same pattern, while stronger models are mostly uninformative because the tasks are too easy.

The broader result is conditional. When the canonical next target has negligible predictability from the post-divergence observable history, any gold scorer credits local behavior only at that guessing probability. Gold-defined severity then approaches a scorer-imposed boundary, and canonical reconvergence does not measure behavioral recovery. Conditional-on-state replay changes the estimand to local continuation quality and produces interior estimates in this suite.

The practical rule is narrower and testable: **before interpreting post-divergence gold agreement, estimate whether the gold target remains reconstructible from what the agent can observe.** If not, report state-conditional competence separately from milestone or final-state success.

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A Aggregate-data sensitivity checks

The aggregate counts in Table 3 are the unique integers consistent with Table 2’ call totals and three-decimal rates. They permit deterministic sensitivity summaries but not task-cluster confidence intervals; those require the task-level release artifact.

B What each scoring regime identifies

Table 3: Aggregate counts and alternative fit-free propagation summaries. $A_d = \bar{p}_d - \bar{g}_d$; $E_d = (1 - \bar{g}_d)/(1 - \bar{p}_d)$.

Model	d	baseline	free-running	A_d	E_d
llama-3.1-8b	1	42/60	42/60	0.000	1.000
	2	83/120	71/120	0.100	1.324
	4	156/260	95/260	0.235	1.587
	6	223/390	70/390	0.392	1.916
	8	100/184	32/184	0.370	1.810
allam-2-7b	1	32/60	32/60	0.000	1.000
	2	46/120	42/120	0.033	1.054
	4	111/260	54/260	0.219	1.383
	6	114/240	36/240	0.325	1.619

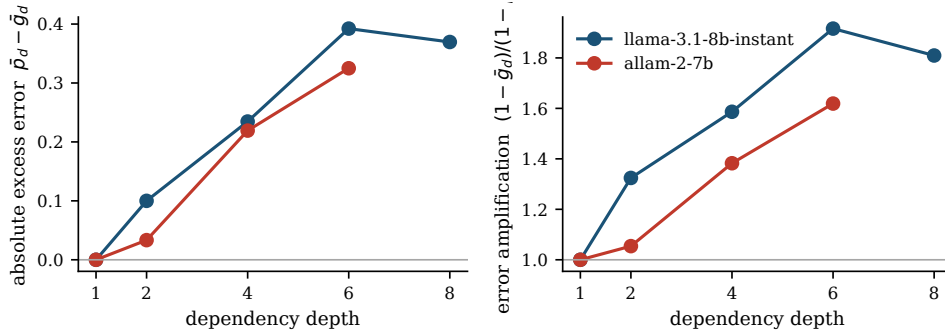


Figure 6: The propagation pattern is not specific to $L_d = 1 - \bar{g}_d/\bar{p}_d$. Absolute excess error grows to 0.392 and 0.325 by depth 6; free-running errors are amplified by factors of 1.916 and 1.619 relative to baseline errors. Both metrics plateau or terminate after depth 6 rather than supporting strict monotonicity.

Table 4: Scoring regimes answer complementary questions.

Scorer	Alternative paths	Primary estimand	Post-divergence limitation
Canonical match	Usually no	Agreement with one reference	Local competence is confounded when the target has small ϵ_k
Conditional-on-state (ours)	Yes, one state at a time	Correct local continuation	Does not imply final task success
Milestone DAG (ToolSandbox)	Yes	Intermediate/final progress	Attribution depends on milestone coverage
Final-state tests (AppWorld, τ -bench)	Yes	End-task state and side effects	Usually lacks per-call diagnosis